NN-H-104
August 11, 1965
SUPERSEDING
See 6.3

FEDERAL SPECIFICATION

HANDLES, WOOD, ACME THREADED AND CIRCULAR TAPERED END

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for wood handles used with brooms, brushes, squeegees, and paint rollers. (See 6.1.)

1.2 Classification.

1.2.1 *Types, sizes, and lengths.* Wood handles furnished under this specification shall be of the following types, sizes, and lengths, as specified (see 6.2):

Types

I—Acme threaded end.
II—Circular tapered end.

Sizes

1—7/8 inch diameter.
2—15/16 inch diameter.
3—1-1/8 inch diameter.

Length (feet)

4
4-1/2
5
6
8
10
12

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-636—Box, Fiberboard.
PPP-T-97—Tape, Pressure-Sensitive Adhesive, Filament Reinforced.
UU-P-268—Paper, Kraft, Untreated, Wrapping.
UU-T-111—Tape, Paper, Gummed (Sealing and Securing).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional C fices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specification:

MIL-W-6110—Wood, Determination of Moisture Content.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—*Marking for Shipment* and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Wood.* Wood shall be a close-grained and straight-grained ash, beech, birch, cherry, elm, hickory, ramin (Gonystylus spp.), pecan, maple, red or white oak, Douglas-fir or southern yellow pine. The wood shall be thoroughly seasoned so that the finished handle when tested in accordance with MIL-W-6110 has a moisture content of not less than 8 percent and not more than 12 percent. The wood shall be free of decay, knots, shakes, wane, wormholes, heart center, and brashness.

3.2 Construction. Handles shall be made of the wood specified in 3.1.1, except that Douglas-fir and southern yellow pine shall not be used for the type I handles. Handles fabricated from woods with a cross grain or spiral grain sloping more than 1-inch in 15 inches shall not be acceptable. The diameter of the handle shall be uniform throughout its length except at the threaded or tapered end and the handle shall be straight. A bow or warp in the length of the handle that causes an eccentricity of not more than 3/8 inch when the handle is rolled on a flat surface, shall be acceptable.

3.2.1 The hand ends of the handles shall be rounded and the brush ends shall be threaded for the type I handles and tapered for the type II handles, as specified (see 1.2 and 6.2).

3.2.1.1 *Threaded ends, type I.* Type I handles shall have an Acme type thread, five threads per inch, for a length of 1-1/16 inches, conforming to the dimensions and tolerances in figure 1. The threads shall be firm, continuous, round and free of chips, gouges and wood imperfections in 3.1.1. Threaded ends may terminate in a square or rounded shoulder.

3.2.1.2 *Tapered ends, type II.* Type II handles shall taper starting 3-inches from the end of the handle, and terminate in a 3/4 inch minor diameter. The taper shall conform to figure 1, and shall be straight, uniform, round, and free of the wood defects in 3.1.1.

3.3 Size. Handles furnished shall have a diameter of 7/8 inch, 15/16 inch, or 1-1/8 inches, as specified (see 6.2). A tolerance of plus or minus 1/32-inch shall apply to all sizes.

3.4 Length. Handles shall be furnished in the length specified (see 1.2.1 and 6.2). A tolerance of plus or minus 1/2 inch shall apply to all handles under 10 feet in length and plus or minus 1-inch shall be permitted for handles 10 feet in length and over. Handles 10 feet in length or over shall be made in two sections of equal length which shall be joined by a quick-disconnect type coupling that does not require the use of tools for assembly or disassembly.

3.5 Finish. Handles shall be sanded smooth throughout except for the threaded or tapered end, and finished with two coats of transparent varnish, lacquer or other clear organic coating. The coating shall be continuous, uniform and smooth, and thick enough to furnish a glossy appearance.

3.6 Workmanship. Handles shall be straight, free of loose or raised grain, and the wood defects indicated in 3.1.1.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Component and material inspection.* In accordance with 4.1, the supplier is responsible for insuring that components and materials used were manufactured, examined and tested in accordance with this specification or referenced subsidiary specifications to the extent specified.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with MIL-STD-105. For purposes of sampling, a lot shall consist of all handles of the same type, size, and length offered for delivery at one time.

4.2.1 *End-item inspection.* Each lot of handles shall be examined in accordance with 4.2.1.1, and tested in accordance with 4.2.1.2. The lot size shall be expressed in units of handles, and the sample unit shall be one handle.

4.2.1.1 *Visual examination.* Each lot shall be examined for defects classified in table I. Inspection level II shall apply and acceptance shall be based on an acceptable quality level (AQL) of 1.5 percent defectives for majors and 6.5 percent defectives for totals, on a percent defectives basis.

TABLE I. *Classification of defects*

Examination	Defect	Classification	
		Major	Minor
Wood	Not species specified		X
	Any decay, shake, heart center, or brashness in the wood	X	
	Not close grained wood		X
	Cross grain or spiral grain sloping more than 1-inch in 15-inches		X
	Crook (warp or bow) more than 3/8 inch when handle is rotated on flat surface		X
	Any knot hole or loose knot	X	
	Any tight or firmly encased knot 1/8 inch or more in diameter		X
	Any season check, fracture, or split, open or visibly concealed with filler	X	
	Any wormhole, larva channel or pocket, open or visible concealed with filler		X
	Any wane (lack of wood)	X	
	Loose, raised or torn grain producing a rough surface	X	
Construction	Not handle type specified	X	
	Not rounded at hand end		X
	Brush end of handle not threaded or tapered as specified	X	
	Handle not uniform diameter throughout its length except at threaded or tapered end		X
	Handle out of round		X

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TABLE I. *Classification of defects (cont'd)*

Examination	Defect	Classification	
		Major	Minor
Construction (cont'd)	Handle 10 foot or more in length not two piece construction with sections of equal length ...	X	
	Coupling not furnished for two piece construction .	X	
Dimensions	Diameter not as specified or not within tolerance specified		X
	Handle length not as specified or not within tolerance specified		X
	Acme thread not within tolerances shown in figure 1 .	X	
	Taper not within tolerances shown in figure 1	X	
Finish	Not lacquered, varnished or coated with clear organic coating	X	
	Coating not glossy .		X
	Wet or tacky film area		X
	Not smoothly sanded		X
	Coating not continuous or uniform		X
	Dirt or foreign matter embedded in finish.		X
Workmanship	Threads not continuous, not firm or free of imperfections	X	
	Taper out of round .		X

4.2.1.2 *Testing of the end item.* Each lot of handles shall be tested in accordance with 4.4 for moisture content of the wood. The sample unit shall be one handle and the sample size for testing shall be selected in accordance with 4.2 at inspection level S-2. Any handle containing less than 8 or more than 12 percent moisture shall be considered defective. The AQLs shall be 4.0 on a percent defective basis.

4.3 Examination of the preparation for delivery. The packaging, packing, and marking of each lot offered for delivery shall be examined to determine compliance with section 5 and the special requirements in the contract or order. The lot size shall be expressed in units of shipping containers and the sample unit shall be one shipping container fully prepared for delivery. Inspection level S-2 shall apply and lot acceptance be determined on the basis of an AQL of 4.0 defects per hundred units. The examination shall be made for the defects in table II.

TABLE II. *Defects in the preparation for delivery*

Examine	Defect
Marking (unit package, shipping container)	Omitted, incomplete, incorrect, illegible.
	Not in accordance with contract requirements.
Packaging	Not unit packaged as specified.
	End trays missing or not secured.
	Less than two bands of filament tape.
	Improper materials used for packaging.
Packing	Not packed as specified.
	Any packing component or material missing, damaged, or not as specified.
	Closure not accomplished by specified or required method or materials
Workmanship	Inadequate application of components or materials such as: incomplete closure of container flap, taping or banding loose or missing; bulging or distortion of the container.
Content	Number of units or intermediate packages not in accordance with requirements.

4.4 Test Methods.

4.4.1 *Moisture content of the wood.* The moisture content of the wood handles shall

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be determined in accordance with MIL-W-6110 using the electrical resistance type moisture meter. The determination shall be made two feet from either end of the handle.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and applications of the various levels of packaging and packing.)

5.1 Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.* Twelve handles of one type, size and length shall be formed into a tight rectangular bundle by applying not less than two bands of pressure-sensitive tape conforming to type III of PPP-T-97 or two ties of twine conforming to type I of T-T-881. The ends of the bundle shall be protected by scored and slotted fiberboard trays extending not less than 6 inches from the ends of the handles. The trays shall be secured in place with filament tape. Alternatively, the bundle may be wrapped with two sheets of 50/60 pound kraft paper conforming to UU-P-268 and secured with gummed kraft tape conforming to UU-T-111.

5.1.2 *Level C.* Handles shall be packaged in accordance with the suppliers' commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Thirty-six handles, packaged in accordance with 5.1, as specified (see 6.2), shall be packed in a style OPF or FPF, weather-resistant, type CF or SF fiberboard box conforming to PPP-B-636. The box shall be closed and reinforced in accordance with the appendix to the box specification.

5.2.2 *Level B.* Handles shall be packed in accordance with 5.2.1 except that the boxes shall be class domestic, variety SW, grade 200, and shall be reinforced with either tape banding or metal straps.

5.2.3 *Level C.* Handles, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest rates. Shipping containers shall be in accordance with rules and regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special markings required by the contract or order, shipments shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special markings specified in the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Handles furnished under this specification are intended for use with brushes, brooms, squeegees and paint rollers under the following specifications:

H-B-71—Brooms, Rattan; Push.
H-B-190—Brushes, Dusting, type II.
H-B-271—Brushes, Garage, Fiber.
H-B-531—Brush, Scrub, Deck.
H-B-651—Brush, Floor, Sweeping.
H-B-751—Brushes, Window.
ZZ-S-666—Squeegees.
H-R-00550—Roller Kit, Paint.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, size, and length required. (See 1.2.)
(*c*) Applicable level of packaging and packing. (See 5.1 and 5.2.)

6.3 Supersession data. This specification supersedes the handle requirements of the following documents, and handles therein contained have the new classification indicated:

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Specification	NN-H-104 classification		
	Type	Size	Length
H-B-71b	II	3	54 inches
H-B-190	I	1	10 feet
H-B-271	I	1	54 inches
H-B-531c	II	3	51 inches
H-B-651e	I	2	60 inches
H-B-751b	II	1	8, 10, 12 feet
ZZ-S-666a	II	1	54 inches

6.4 **Transportation description.** The transportation description and minimum weights applicable to this commodity are:

Rail:

Handles, wooden not otherwise indexed by name, finished.
Carload minimum weight 36,000 pounds.

Motor:

Handles, wooden not otherwise indexed.
Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

CUSTODIANS:

Army—GL
Navy—SA
Air Force—69

Review activities:

Army—GL
Navy—SA
Air Force—69

User activities:

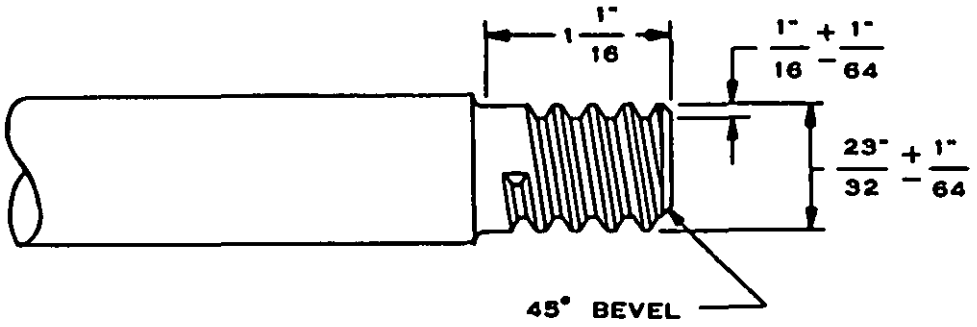
Navy—MC
Air Force—

Preparing Activity:

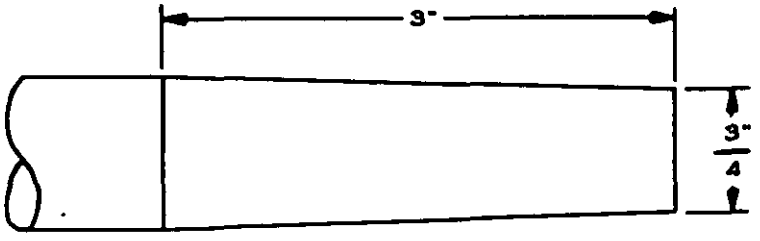
Navy—SA

Review and User information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

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TYPE I-ACME THREADED END



TYPE II-TAPERED END

FIGURE 1. Handles, wood, Acme threaded and tapered end

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